BTEC Level 3 Applied Science · Unit 1 Chemistry · Worksheet

Atomic Structure

Original JDScience practice. These questions were written for this resource and are not official exam-board items.

Learning focus: particles, nuclide notation, isotopes, A_r and ions.

A Retrieval

1. State the relative charge and relative mass of a proton, a neutron and an electron.

2. Define atomic number (Z) and mass number (A).

3. What is the same, and what is different, in a pair of isotopes?

B Knowledge

4. A neutral atom of ^{39}K has atomic number 19. Calculate the numbers of protons, neutrons and electrons.

Working

5. Explain why atoms are electrically neutral even though they contain charged particles.

C Application and calculation

6. Complete the table for ^{24}Mg, $^{24}Mg^{2+}$, ^{19}F and $^{19}F^{-}$.

Species	Protons	Neutrons	Electrons

7. Gallium occurs as 60.0% ^{69}Ga and 40.0% ^{71}Ga. Calculate A_r. Show substitution.

Working

8. Explain why the A_r of chlorine on the periodic table is not a whole number.

D Exam-style practice

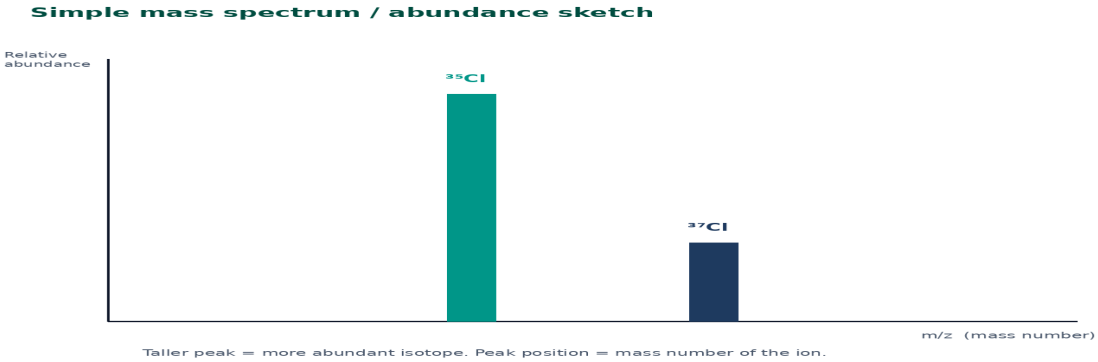
9. A student claims that ^{14}C and ^{14}N are isotopes of each other. Explain why this is incorrect. [2]

10. Calculate A_r for an element that is 75% isotope-63 and 25% isotope-65. [2]

Working

11. Explain how a 2– ion forms from a neutral atom, referring to protons and electrons. [3]

E Challenge

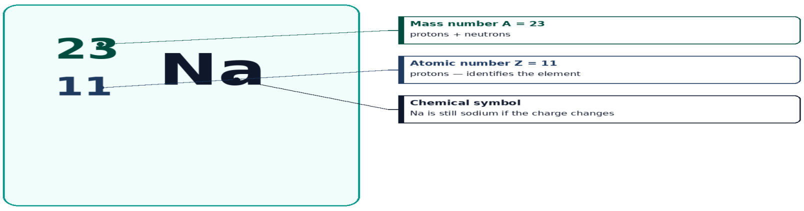


12. Neon has peaks at mass numbers 20, 21 and 22 with relative abundances 90.5, 0.27 and 9.25. Calculate A_r to 3 s.f. and comment on the effect of ignoring the middle isotope.

Working

13. On the nuclide diagram below, label A, Z, protons and neutrons for $^{23}_{11}Na$. Then state the electron number of Na^{+}.

Nuclide notation · sodium-23



Neutrons = A − Z = 23 − 11 = 12

Neutral atom: electrons = Z = 11

Na^{+} ion: still 11 protons; electrons = 10. The nucleus does not change.